RozarNLP-cr7 at WMT26 Indic Machine Translation Shared Task

Abstract

This paper describes the submission of Team RozarNLP-cr7 to the WMT26 Indic Machine Translation Shared Task. We focus on machine translation between English and several low-resource languages from Northeast India, including Assamese, Manipuri, Meitei, Mizo, Khasi, and Nyishi. Our work investigates the effectiveness of multilingual transfer learning and parameter-efficient adaptation for low-resource machine translation.

Our primary systems are based on NLLB-200 Distilled 1.3B adapted using Low-Rank Adaptation (LoRA). To explore multilingual transfer across related language pairs, we train two multilingual models using only the official shared-task training data. The first model is jointly fine-tuned for Assamese, Manipuri (Bengali script), and Mizo translation in both directions, while the second model is jointly fine-tuned for Khasi and Nyishi. During experimentation, we observed challenges in adapting Manipuri written in Meitei script within the multilingual setting. To further investigate this behavior, we additionally investigate adaptation of Manipuri written in Meitei Mayek script. While multilingual LoRA adaptation proved effective for Assamese, Manipuri (Bengali script), Mizo, Khasi, and Nyishi, adaptation to Meitei Mayek remained challenging, and the pretrained NLLB-200 system was retained for this language pair.

As contrastive systems, we evaluate pretrained IndicTrans2 and NLLB-200 models without task-specific adaptation. Our approach directly addresses key objectives of the shared task by leveraging multilingual training, transfer learning from large pretrained multilingual models, and parameter-efficient fine-tuning for low-resource languages.

Preliminary validation experiments indicate that multilingual LoRA adaptation improves performance over the corresponding pretrained baselines for most language pairs. The findings highlight the effectiveness of multilingual transfer for low-resource Indic machine translation and provide insights into the challenges associated with script-specific adaptation in multilingual translation systems.

1. Introduction

Machine Translation (MT) for low-resource Indic languages remains challenging due to limited parallel corpora, domain diversity, and scarce linguistic resources. Recent multilingual pretrained models have demonstrated strong transfer capabilities across languages, making them attractive solutions for low-resource translation tasks.

The WMT26 Indic Machine Translation Shared Task provides an opportunity to evaluate multilingual neural machine translation systems on several underrepresented languages from the Indian subcontinent. Our submission explores the effectiveness of pretrained multilingual translation models and lightweight adaptation strategies for improving translation quality under limited-resource settings.

2. Data

Only the official datasets released as part of the WMT26 Indic Machine Translation Shared Task were used for training and adaptation. No additional parallel corpora were included in the submitted primary systems. The evaluated language pairs are:

- English ↔ Assamese
- English ↔ Manipuri (Meitei + Bengali)
- English ↔ Mizo
- English ↔ Khasi
- English ↔ Nyishi

Note: Manipuri and Meitei refer to the same language — Meitei (also called Meitilon) is the official name for the language spoken predominantly in Manipur. Both labels appear in the shared task following the organizers' naming conventions. Bidirectional training examples are created by including both source-to-target and target-to-source translation directions. The official test sets supplied by the organizers are used exclusively for generating submission outputs.

3. System Description

Language Pair	Primary	Contrastive
Assamese	NLLB+LoRA	IndicTrans2
Manipuri	NLLB+LoRA	IndicTrans2
Meitei	NLLB Base	No
Mizo	NLLB+LoRA	NLLB Base
Khasi	NLLB+LoRA	No
Nyishi	NLLB+LoRA	No model supports

3.1 Submission Overview

Our submission consists of two multilingual NLLB-200 Distilled 1.3B models adapted using Low-Rank Adaptation (LoRA). All submitted systems are trained exclusively on the official training data released by the WMT26 Indic Machine Translation Shared Task organizers.

Primary System (Constrained): The primary submission consists of two LoRA-adapted NLLB-200 models:

- Model A: Assamese–Manipuri (Bengali script)–Mizo Multilingual Model (English ↔ Assamese, Manipuri, Mizo — both directions)
- Model B: Khasi–Nyishi Multilingual Model (English ↔ Khasi, Nyishi — both directions)

Both models share parameters across multiple language pairs to improve multilingual transfer while maintaining a small number of trainable parameters through LoRA adaptation.

Contrastive System: The contrastive submission uses pretrained baseline systems without task-specific adaptation:

- IndicTrans2 for Assamese and Manipuri/Meitei.
- NLLB-200 Distilled 1.3B (no LoRA adaptation) for Mizo and Khasi

These systems provide a direct comparison between pretrained multilingual models and the LoRA-adapted systems used in the primary submission.

3.2 Assamese–Manipuri/Meitei–Mizo Multilingual Model

The Assamese–Manipuri/Meitei–Mizo multilingual model combines four language pairs within a single LoRA-adapted NLLB-200 model. Assamese (asm_Beng) and Manipuri (mni_Beng) are directly supported by the pretrained NLLB-200 vocabulary. Additional language identifiers were incorporated where required to support the remaining language pairs and scripts used in our experiments. This multilingual setup enables parameter sharing across related low-resource translation tasks while maintaining a compact adaptation footprint through LoRA.

Manipuri (Meitei Script) Adaptation

Preliminary experiments showed that multilingual adaptation was effective for Assamese, Mizo, and Manipuri written in Bengali script (mni_Beng). However, substantially weaker performance was observed for Manipuri written in Meitei script (mni_Mtei), including frequent empty outputs and unstable generations.

During preliminary experiments, we investigated adaptation of Manipuri written in Meitei Mayek (mni_Mtei). Although a custom language identifier was introduced and multilingual LoRA adaptation was attempted, validation performance remained poor, with frequent empty outputs and unstable generations. Because of these difficulties, the submitted system for the Meitei language pair uses the pretrained NLLB-200 model without additional adaptation.

Preliminary experiments suggest that script-specific adaptation for Meitei Mayek remains an open challenge and warrants further investigation.

3.3 Khasi–Nyishi Multilingual Model

The second multilingual model is based on NLLB-200 Distilled 1.3B and is jointly trained on Khasi and Nyishi parallel corpora. Khasi is natively supported through the FLORES-200 language inventory (language code kha_Latn). Nyishi, which is not part of the standard FLORES-200 inventory, is incorporated using a custom language identifier (nif_Latn) registered via the slow NllbTokenizer to ensure compatibility during tokenization. Both translation directions are included during training to maximize multilingual transfer. A shared multilingual LoRA adapter enables transfer learning between the two low-resource language pairs while keeping the number of trainable parameters small.

The language pairs were grouped into two multilingual models based on language support, script characteristics, and practical adaptation considerations. Assamese and Manipuri are represented in the Bengali script, while Meitei and Mizo were jointly adapted within the same multilingual model to encourage transfer learning across multiple low-resource language pairs. Khasi and Nyishi were trained in a separate multilingual model because both languages use the Latin script and represent highly resource-constrained translation tasks.

4. Training Configuration

Component	Configuration
Base Model	NLLB-200 Distilled 1.3B (~1.3B parameters)
Fine-Tuning Method	Low-Rank Adaptation (LoRA)
LoRA Rank (r)	16
LoRA Alpha	32
LoRA Dropout	0.05
Target Modules	q_proj, k_proj, v_proj, out_proj
Optimizer	AdamW
Learning Rate	2×10^{-4}
LR Scheduler	Cosine Decay
Warmup Ratio	0.06
Gradient Accumulation Steps	8

Effective Batch Size	32
Number of Epochs	2
Precision	BF16 / FP16
Framework	PyTorch + Hugging Face Transformers + PEFT
Hardware	NVIDIA RTX 2000 Ada Generation Laptop GPU (16GB VRAM)

5. Inference

For NLLB-based systems, translation is generated using autoregressive decoding with explicit target-language conditioning through language-specific decoder initialization. For IndicTrans2-based systems, translations are generated using the official inference pipeline with language-specific preprocessing and postprocessing. Translations are generated in the original order of the provided test sets. Each submission file contains one translated sentence per line corresponding to the official evaluation dataset.

6. Preliminary Evaluation

We conducted preliminary validation experiments to compare pretrained multilingual translation models with the proposed LoRA-adapted NLLB-200 systems. Evaluation used BLEU and chrF as automatic metrics.

6.1 Assamese–Manipuri(Bengali) –Mizo Multilingual Model

System	Language	BLEU	chrF
IndicTrans2 Baseline	Assamese	9.16	39.32
IndicTrans2 Baseline	Manipuri / Meitei	2.82 / 1.74	39.79 / 37.45
IndicTrans2 Baseline	Mizo	0.16	0.44
NLLB-200 Baseline	Mizo	17.47	44.617
NLLB-200 + LoRA (Multilingual)	Assamese	33.65(as -> en) 17.01(en->as)	53.05(as->en) 43.02(en->as)
NLLB-200 + LoRA (Multilingual)	Manipuri (Bengali)	34.87(mni-en) 15.92(en-mni)	60.70(mni-en) 55.02(en-mni)
NLLB-200 + LoRA (Multilingual)	Mizo	37.51(lus-en) 22.03(en-lus)	55.57(lus-en) 46.72(en-lus)

NLLB-200 +LoRA (Multilingual)	Manipuri(Meitei)		

Note: The reported BLEU and chrF scores were computed on held-out validation data and are shown separately for each translation direction.

6.2 Khasi–Nyishi Multilingual Model

System	BLEU	chrF
NLLB-200 Base (no adaptation)	0.0	0.0
NLLB-200 + LoRA	14.63	12.61

The NLLB-200 Base contrastive system for Khasi and Nyishi serves as a reference comparison in the final submission. The multilingual Khasi–Nyishi LoRA adaptation demonstrates the feasibility of parameter-efficient adaptation for extremely low-resource language pairs using only the official shared-task training data. Baseline contrastive scores will be reported in the final system description.

7. Reproducibility

Source code, training scripts, inference scripts, and preprocessing utilities are publicly available at:

https://github.com/Iamdevsuyash/WMT26

The repository contains data preparation scripts, training pipelines, LoRA adaptation scripts, inference utilities, evaluation scripts, and configuration files. Datasets are not redistributed and must be obtained from their original sources.

All experiments were conducted using only the official WMT26 shared-task training data. No external parallel corpora were used in the submitted primary systems.

8. Conclusion

This work investigates multilingual transfer learning and parameter-efficient adaptation for low-resource Indic machine translation. We evaluate multilingual pretrained translation models — including IndicTrans2 and NLLB-200 — and apply LoRA-based adaptation across six Northeast Indian language pairs. Our experiments demonstrate the practicality of adapting multilingual pretrained models for extremely low-resource settings while maintaining modest computational requirements, achieving substantial BLEU gains over pretrained baselines for the languages evaluated.

9. Future Work

Future directions include:

- Per-language LoRA adapters for Khasi and Nyishi to reduce cross-language interference.
- Back-translation for synthetic data augmentation, which was not pursued in this submission due to time and compute constraints.
- Continued multilingual adaptation using monolingual data resources.
- Domain adaptation to improve robustness across different text genres.
- Multilingual joint training across related Northeast Indian languages.
- Investigation of larger translation models and more advanced parameter-efficient adaptation methods.
- Future work will investigate the trade-off between multilingual and language-specific adaptation for low-resource languages, particularly for language pairs requiring newly introduced script-specific language identifiers.